

Cold Chain Logistics Performance in Saudi Arabia: Evidence from the Pharmaceutical Sector

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Course: OSCM411
Focus companies: Tamer Group, Medical Fakeeh (Saudi Airlines Medical Services), others

What is Cold Chain Logistics?

A temperature-controlled supply chain. Critical for pharmaceutical products such as:

-Vaccines
-Insulin
-Biologics
-Temperature-sensitive medicines



Why it matters?

Temperature deviations can

-Reduce drug effectiveness
-Cause financial losses
-Risk patient safety

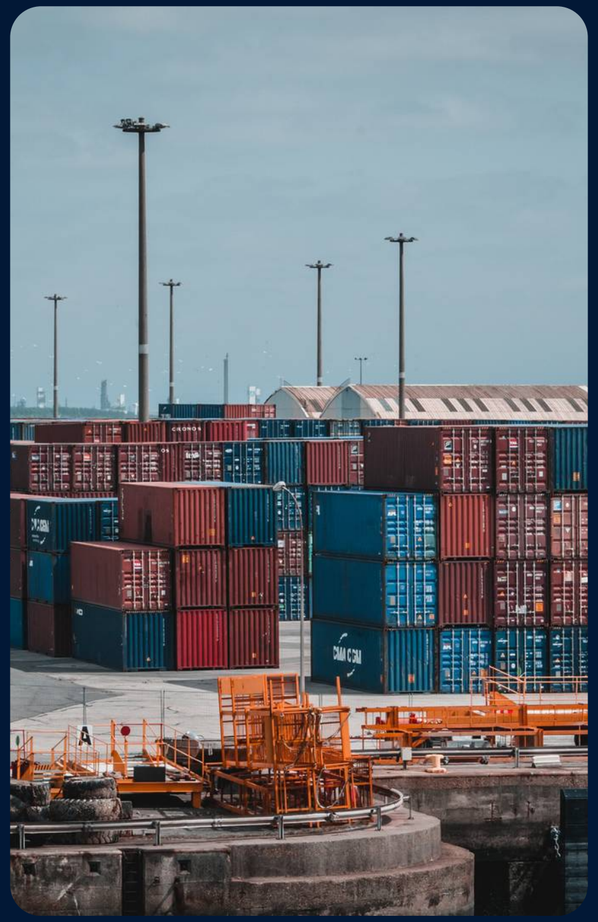
Research Problem

Saudi Arabia has:

Extreme climate conditions

Long transportation distances

Growing pharmaceutical demand



Research Question

How effective are cold chain logistics practices in Saudi Arabia's pharmaceutical sector?

Yet limited empirical evidence on how effective cold chain practices actually are

Importance to Operations & Supply Chain Management

This research contributes to OSCM by examining:

- Logistics performance
- Risk management
- Practical Importance
- Quality control
- Technology adoption







This research supports:

- Pharma distributors
- Hospitals
- Regulators (SFDA)
- Aligns with Vision 2030 healthcare transformation



Research Methodology

Research Design

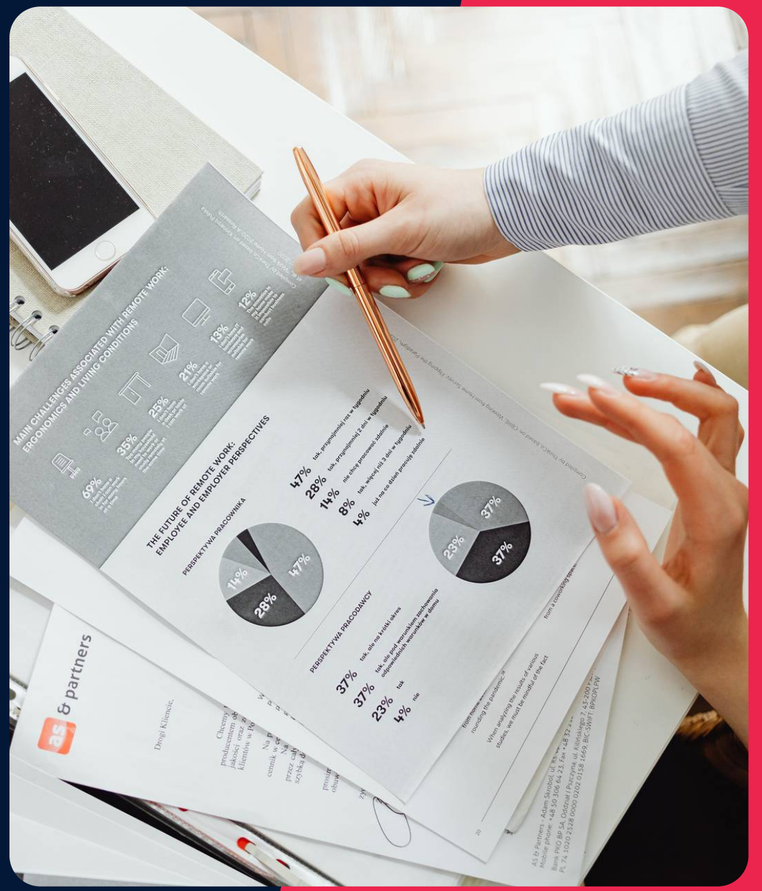
Qualitative + descriptive approach

Data Collection Methods

1. Interviews with:
   - Logistics managers
   - Quality assurance officers
2. Secondary data:
   - Cold chain policies
   - SOPs
   - Temperature records (where accessible)

Target Organizations

1. Tamer Group
2. Medical Fakeeh / Saudi Airlines Medical Services
3. Other major pharma logistics players





فقيه.
ميديكال فقيه
Medical Fakeeh

TAMER

Expected Insights & Challenges

Expected Findings

- Use of advanced cold chain technologies
- Strong compliance in large firms
- Growing reliance on data loggers and monitoring systems

Key Challenges

- Temperature excursions during last-mile delivery
- Human handling errors
- Cost of advanced monitoring systems
- Harsh climate conditions

Link to OSCM:

- Trade-off between cost efficiency and service quality



Conclusion

Cold chain logistics is critical for pharmaceutical supply chains.

Effective monitoring supports:

- Product integrity
- Regulatory compliance
- Patient safety



Research Contribution

- Provides local evidence from Saudi Arabia
- Helps organizations improve cold chain reliability
- Supports better healthcare outcomes
- Practical insights for Saudi pharma logistics
- A foundation for future OSCM research


Thank you!